

Week 3

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1. Quick Recap

Euclidean Distance between Two Vectors

Let $u, v \in \mathbb{R}^d$. The euclidean distance between u and v is given by $||u - v||$ or $||v - u||$.

Linear Combination

Let V be a vector space. Let $v_1, v_2, \dots, v_n \in V$ and let $a_1, a_2, \dots, a_n \in \mathbb{R}$.

$$z = a_1v_1 + a_2v_2 + \dots + a_nv_n = \sum_{i=1}^n a_i \cdot v_i$$

z is a **linear combination** of $v_1, v_2, \dots, v_n \in V$.

Span Set

Let $S = \{v_1, v_2, \dots, v_n\}$ be a set of vectors in $\mathbb{R}^d$ where $d \in \mathbb{R}$.

The span set of S is defined as follows.

$$\text{span}(S) = \left\{ \sum_{i=1}^n a_i \cdot v_i \mid v_i \in S \text{ and } a_i \in \mathbb{R} \right\}$$

In simple words, $\text{span}(S)$ is the set of all possible linear combinations of $v_1, v_2, \dots, v_n$.

NOTE : $\text{span}(S)$ is a vector space.

Linear independence and Linear dependence.

Let $S = \{v_1, v_2, \dots, v_n\}$ be a set of vectors in $\mathbb{R}^d$ where $d \in \mathbb{R}$.

Consider the following system

$$\begin{bmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

- If this system has a **unique solution**, then S is a **linearly independent** set.
- If this system has **infinitely many solutions**, the S is a **linearly dependent** set.

In simple words, if there exists **only** one way to linearly combine the vectors in set S to get $\begin{bmatrix} 0 & 0 & \cdots & 0 \end{bmatrix}^T$, then S is **linearly independent**. If there are more than one way to linearly combine the vectors in set S to get $\begin{bmatrix} 0 & 0 & \cdots & 0 \end{bmatrix}^T$, then S is **linearly dependent**.

NOTE : If set S is linearly independent, then $\begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}^T = \begin{bmatrix} 0 & 0 & \cdots & 0 \end{bmatrix}^T$. This is because, $\begin{bmatrix} 0 & 0 & \cdots & 0 \end{bmatrix}^T$ is always a solution to a homogeneous system of linear equations.

Basis and Dimension of a Vector Space

Let V be a vector space. Let B be a subset of V . B is a basis of V if it satisfies the following conditions.

- B is a linearly independent set
- $\text{span}(B) = V$

If B is a basis of V , then the **dimension** of V is the cardinality of B . In simple words, the **dimension** of V is the number of elements in the basis of V .

The dimension of a vector space is unique. The basis set is not unique.

Rank of a Matrix

Let A be an $m \times n$ matrix.

The rank of A is the number of linearly independent columns of A

- Column space of $A \rightarrow \text{span}\{\text{All the column vectors of } A\}$
- Column rank of $A \rightarrow \dim(\text{Column space of } A)$.
- Row space of $A \rightarrow \text{span}\{\text{All the row vectors of } A\}$
- Row rank of $A \rightarrow \dim(\text{Row space of } A)$

Nullity of a Matrix

Let A be an $m \times n$ matrix.

The set of all the solutions for $Ax = 0$ is the nullspace of A .

$$\text{Nullspace}(A) = \{x | Ax = 0\}$$

The dimension of the $\text{Nullspace}(A)$ is the nullity of A .

2. Few Facts

Let A be a matrix. There are two facts mentioned below, that are related A . Though the proofs for these two are not given, both are accompanied by small hints which are useful to prove these two facts. Readers are requested to revisit the concepts **Maths-2** for a complete proof.

$$\text{Row Rank}(A) = \text{Column Rank}(A)$$

HINT

Let R be the reduced row echelon form of A .

- The row rank of A is the number of non-zero rows in R .
- The column rank of A is the number of pivot columns in R .

The **number of non-zero rows** in R and the **number of pivot columns** in R are equal.

$$\text{Rank}(A) + \text{Nullity}(A) = \text{No. of columns in } A$$

HINT

For the system of linear equations denoted by $Ax = 0$, let $(A|0)$ be the augmented matrix. Let $(R|0)$ be the reduced row echelon form of $(A|0)$.

- The number of dependent variables is equal to the rank of A
- The number of independent variables is equal to the nullity of A .

The **sum** of the **number of independent variables** and the **number of dependent variables** is equal to the **number of columns** in A .

3. Orthogonality

Two vectors $u, v \in \mathbb{R}^d$ are orthogonal to each other if $u \cdot v = 0$. If the dot product of two vectors is 0, then the angle between them is 90° .

Orthogonal Set

A set $V \in \mathbb{R}^d$ is an orthogonal set, if every pair of distinct vectors in V is orthogonal.

Theorem : An orthogonal set (without the zero vector) is a linearly independent set

Proof: Let $V = \{v_1, v_2, \dots, v_n\}$ be an orthogonal set. Let $c_i \in \mathbb{R}, i \in \{1, \dots, n\}$

$$\begin{aligned}c_1 v_1 + c_2 v_2 + \dots + c_n v_n &= 0 \\v_1 \cdot [c_1 v_1 + c_2 v_2 + \dots + c_n v_n] &= 0 \cdot v_1 \\c_1(v_1 \cdot v_1) + c_2(v_1 \cdot v_2) + \dots + c_n(v_1 \cdot v_n) &= 0 \\c_1 \times ||v_1||^2 &= 0 \\c_1 &= 0\end{aligned}$$

Similarly, we can show that $c_i = 0, \forall i \in \{1, \dots, n\}$. Therefore, V is a linearly independent set.

Orthonormal Set

A set $V \in \mathbb{R}^d$ is an orthonormal set, if the following conditions are true:

- V is an orthogonal set
- $||u|| = 1, \forall u \in V$

Orthogonal Subspace

Two subspaces $U, V \in \mathbb{R}^d$ are orthogonal subspaces if every vector in U is orthogonal to every vector in V and vice-versa. In other words, if U and V are orthogonal subspaces, then,

$$u \cdot v = 0, \forall u \in U \text{ \& \& } \forall v \in V$$

Examples:

- The XY – plane and the Z axis are orthogonal subspaces in $\mathbb{R}^3$.
- The X – axis and the Y – axis are orthogonal subspace in $\mathbb{R}^2$

Orthogonal Complements

Let V be a vector space. Two orthogonal subspaces of V , given by M & N are orthogonal complements, if any one of the conditions given below hold true:

- $\dim(M) + \dim(N) = \dim(V)$
- $V = \text{span}(\text{Basis}(M) \cup \text{Basis}(N))$
- $N = \left\{ x \in V \mid m \cdot x = 0, \forall m \in M \right\}$
- $M = \left\{ x \in V \mid n \cdot x = 0, \forall n \in N \right\}$

In fact, these four conditions are equivalent.

Examples:

- The XY – plane and the Z axis are orthogonal complements in $\mathbb{R}^3$.
 - The X – axis and the Y – axis are orthogonal complements in $\mathbb{R}^2$
-

4. Four Fundamental Subspaces

Consider matrix $A_{m \times n}$. (m rows & n columns). Let us denote each row as r_i , $i \in \{1, \dots, m\}$ and each column as c_j , $j \in \{1, \dots, n\}$.

$$A = \begin{bmatrix} - & r_1 & - \\ - & r_2 & - \\ & \vdots & \\ - & r_m & - \end{bmatrix}_{m \times n}$$

$$A = \begin{bmatrix} | & | & \cdots & | \\ c_1 & c_2 & & c_n \\ | & | & & | \end{bmatrix}_{m \times n}$$

$$A^T = \begin{bmatrix} | & | & & | \\ r_1 & r_2 & \cdots & r_m \\ | & | & & | \end{bmatrix}_{n \times m}$$

$$A^T = \begin{bmatrix} - & c_1 & - \\ - & c_2 & - \\ & \vdots & \\ - & c_n & - \end{bmatrix}_{n \times m}$$

A^T has n rows and m columns. The rows of A are the columns of A^T and the columns of A are the rows of A^T .

Row Space of A

The row space of A , denoted by $R(A)$, is given by the following set.

$$R(A) = \text{span}(\{r_1, r_2, \dots, r_m\})$$

$$\rightarrow R(A) \subseteq \mathbb{R}^n$$

$\rightarrow \dim(R(A))$ is called the **row rank** of A

Column Space of A

The column space of A , denoted by $C(A)$, is given by the following set.

$$C(A) = \text{span}(\{c_1, c_2, \dots, c_n\})$$

$$\rightarrow C(A) \subseteq \mathbb{R}^m$$

$\rightarrow \dim(C(A))$ is called the **column rank** of A

Null Space of A

The null space of A , denoted by $N(A)$, is given by the following set.

$$N(A) = \left\{ x \in \mathbb{R}^n \mid Ax = 0 \right\}$$

$$\rightarrow N(A) \subseteq \mathbb{R}^n$$

$\rightarrow \dim(N(A))$ is called the **nullity** of A

If a vector $v \in N(A)$, then $r_i \cdot v = 0, \forall i \in \{1, 2, \dots, m\}$. That is, every vector in the null space of A is orthogonal to every row vector of A , and, vice-versa.

$$Ax = 0$$

$\downarrow$

$$\begin{bmatrix} - & r_1 & - \\ - & r_2 & - \\ & \vdots & \\ - & r_m & - \end{bmatrix} \times \begin{bmatrix} | \\ x \\ | \end{bmatrix} = \begin{bmatrix} r_1 \cdot x \\ r_2 \cdot x \\ \vdots \\ r_m \cdot x \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Clearly, $R(A)$ and $N(A)$ are orthogonal subspaces of $\mathbb{R}^n$.

$$\begin{aligned} \text{Rank}(A) + \text{Nullity}(A) &= \text{No. of columns in } A \\ \text{Rank}(A) + \text{Nullity}(A) &= n \end{aligned}$$

We know that $\text{Rank}(A) = \dim(R(A))$. Therefore, $R(A)$ and $N(A)$ are also orthogonal complements of each other.

Null Space of A^T

The null space of A^T , also known as the **left null space** of A , denoted by $N(A^T)$, is given by the following set.

$$N(A^T) = \left\{ x \in \mathbb{R}^m \mid A^T x = 0 \right\}$$

$$\rightarrow N(A^T) \subseteq \mathbb{R}^m$$

$$\rightarrow \dim(N(A^T)) \text{ is called the } \mathbf{nullity} \text{ of } A^T$$

If a vector $v \in N(A^T)$, then $c_j \cdot v = 0, \forall j \in \{1, 2, \dots, n\}$. That is, every vector in the left null space of A is orthogonal to every column vector of A , and, vice-versa.

$$A^T x = 0$$

↓

$$\begin{bmatrix} - & c_1 & - \\ - & c_2 & - \\ & \vdots & \\ - & c_n & - \end{bmatrix} \times \begin{bmatrix} | \\ x \\ | \end{bmatrix} = \begin{bmatrix} c_1 \cdot x \\ c_2 \cdot x \\ \vdots \\ c_n \cdot x \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Clearly $C(A)$ and $N(A^T)$ are orthogonal subspaces of $\mathbb{R}^m$.

$$\text{Rank}(A^T) + \text{Nullity}(A^T) = \text{No. of columns in } A^T$$

$$\text{Rank}(A^T) + \text{Nullity}(A^T) = m$$

We know that $\text{Rank}(A^T) = \dim(R(A^T)) = \dim(C(A))$. Therefore, $C(A)$ and $N(A^T)$ are orthogonal complements of each other.

5. Few Proofs

Before we look at projections, let's look at a few theorems and their proofs. In this section, I is an identity matrix.

Theorem

The inverse of an invertible matrix is unique.

Proof:

Let A be an invertible matrix. Let B & C be matrices such that: $AB = BA = I$ and $AC = CA = I$.

$$\begin{aligned} B &= B \\ &= BI \\ &= BAC \\ &= IC \\ &= C \end{aligned}$$

Since $B = C$, the inverse of an invertible matrix is unique.

Theorem

If S is an invertible and symmetric matrix, i.e., $S^T = S$, then S^{-1} is also symmetric.

Proof:

$$\begin{aligned} S^{-1}S &= I \\ (S^{-1}S)^T &= I^T \\ S^T(S^{-1})^T &= I \\ S(S^{-1})^T &= I \\ S^{-1}S(S^{-1})^T &= S^{-1}I \\ I(S^{-1})^T &= S^{-1} \\ (S^{-1})^T &= S^{-1} \end{aligned}$$

Theorem

The nullspace of $A^T A$ and A are the same.

Proof:

To prove this, we have to show that every element in $N(A)$ is also an element in $N(A^T A)$ and every element in $N(A^T A)$ is also an element in $N(A)$.

Let $v \in N(A)$. Then,

$$\begin{aligned}Av &= 0 \\ A^T Av &= A^T 0 \\ A^T Av &= 0\end{aligned}$$

Therefore, $v \in N(A^T A)$

Let $u \in N(A^T A)$. Then,

$$\begin{aligned}A^T Au &= 0 \\ u^T A^T Au &= 0 \\ (Au)^T Au &= 0 \\ ||Au||^2 &= 0 \\ Au &= 0\end{aligned}$$

Therefore, $u \in N(A)$.

6. Projection

Consider a vector space $V \in \mathbb{R}^d$ and a vector $b \in \mathbb{R}^d$. The goal of projection is to find a vector in V that is closest to b . Among all the vectors in V , the vector which is closest to b is called the **projecton** of b onto V .

We shall first visualize projection onto a line in $\mathbb{R}^2$ and projection onto a plane in $\mathbb{R}^3$ and then generalize the idea of projection.

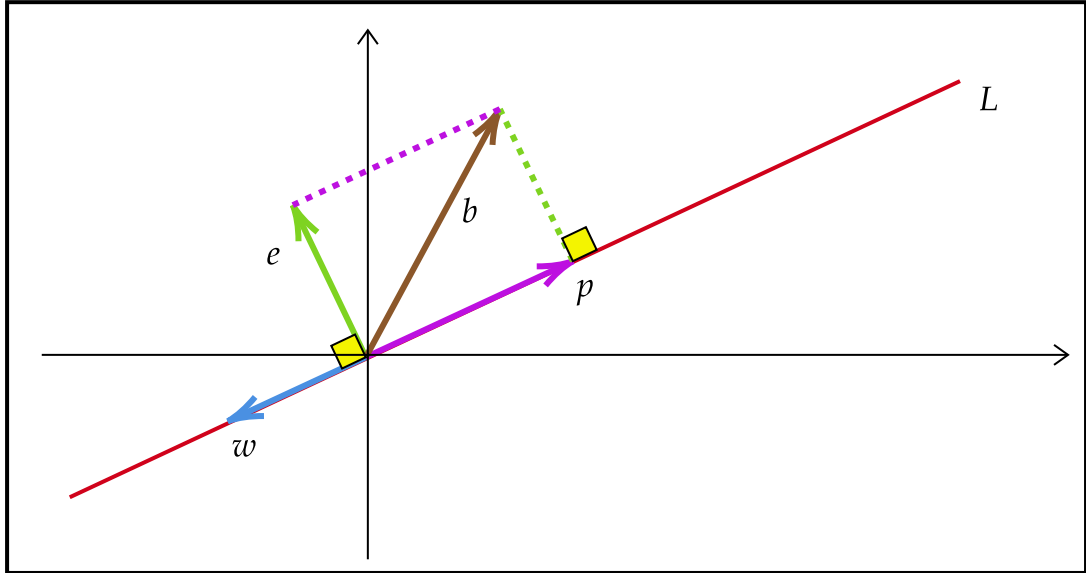
7. Projection onto a line in $\mathbb{R}^2$

Let $b, w \in \mathbb{R}^2$. A line L is given as follows:

$$L = \text{span}(\{w\})$$

Clearly, L is a subspace in $\mathbb{R}^2$.

GOAL : Find the projection of b onto L .



Let us refer to the projection of b onto L as p . Since $p \in L$, p will be a scalar multiple of w . We can write $p = cw$, where $c \in \mathbb{R}$

Among all the vectors in L , p is the vector that is closest to b . The shortest distance is also the perpendicular distance. Therefore, the vector $e = b - p$ is perpendicular to the line L .

The figure shows that $e + p = b$. This is because of the parallelogram law of addition. The vector e is called the **error** vector. The length of this vector gives the distance between b and p .

Since e is perpendicular to the line L , we can write

$$\begin{aligned} w \cdot e &= 0 \\ w \cdot (b - p) &= 0 \\ w \cdot (b - cw) &= 0 \\ w \cdot b - c||w||^2 &= 0 \\ c &= \frac{w \cdot b}{||w||^2} \end{aligned}$$

Once we find c , find p is simple, as $p = cw$.

8. Projection onto a plane in $\mathbb{R}^3$

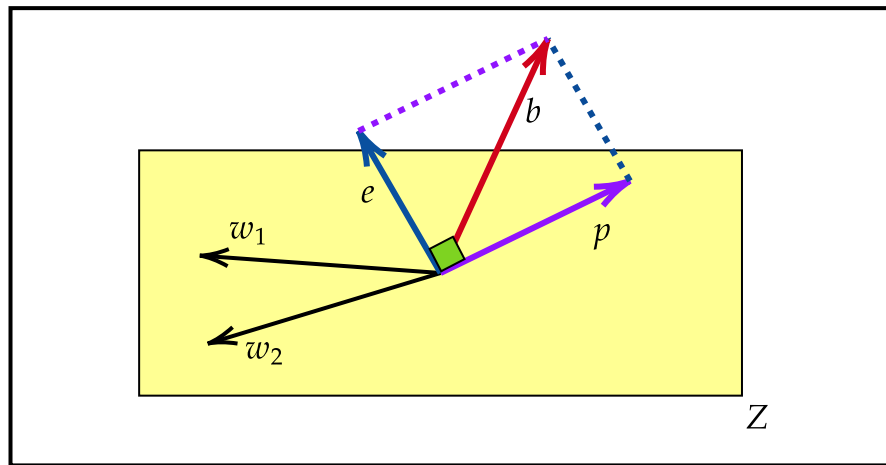
Let $w_1, w_2, b \in \mathbb{R}^3$. Also, let w_1 and w_2 be linearly independent.

A plane Z in $\mathbb{R}^3$ is given as follows:

$$Z = \text{span}(\{w_1, w_2\})$$

Clearly, Z is a subspace of $\mathbb{R}^3$.

GOAL : Find the projection of b onto L



Let us refer to the projection of b onto Z as p . Since $p \in Z$, p will be a linear combination of w_1 and w_2 . We can write $p = \alpha w_1 + \beta w_2$, where $\alpha, \beta \in \mathbb{R}$.

Among all the vectors in Z , p is the closest to b . Again, the shortest distance is the perpendicular distance. Therefore, $e = b - p$ is perpendicular to the plane Z .

Once again, the picture above shows that the parallelogram law of addition holds true for $e + p = b$. Here too, e is called the **error vector**, whose length captures the distance between b and p .

Since e is perpendicular to the plane Z we can write,

$$w_1 \cdot e = 0$$

$$w_1 \cdot (b - p) = 0$$

$$w_1 \cdot (b - \alpha w_1 - \beta w_2) = 0$$

$$\begin{aligned}
w_1 \cdot b - \alpha(w_1 \cdot w_1) - \beta(w_1 \cdot w_2) &= 0 \\
\alpha(w_1 \cdot w_1) + \beta(w_1 \cdot w_2) &= w_1 \cdot b \quad (1)
\end{aligned}$$

$$\begin{aligned}
w_2 \cdot e &= 0 \\
w_2 \cdot (b - p) &= 0 \\
w_2 \cdot (b - \alpha w_1 - \beta w_2) &= 0 \\
w_2 \cdot b - \alpha(w_2 \cdot w_1) - \beta(w_2 \cdot w_2) &= 0 \\
\alpha(w_2 \cdot w_1) + \beta(w_2 \cdot w_2) &= w_2 \cdot b \quad (2)
\end{aligned}$$

The matrix for of linear equations (1) and (2) is $Px = q$, where

$$P = \begin{bmatrix} w_1 \cdot w_1 & w_1 \cdot w_2 \\ w_2 \cdot w_1 & w_2 \cdot w_2 \end{bmatrix}, \quad x = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}, \quad q = \begin{bmatrix} w_1 \cdot b \\ w_2 \cdot b \end{bmatrix}$$

Notice that P is of the form $A^T A$, where $A = \begin{bmatrix} | & | \\ w_1 & w_2 \\ | & | \end{bmatrix}$. The columns of A are linearly independent. As a result, $N(A) = \{0\}$. This means, $N(A^T A) = \{0\}$. Therefore, $A^T A$ is invertible.

$Px = q$ will have a unique solution.

$$\boxed{x = P^{-1}q}$$

9. Projection onto a Subspace

Let $V \subseteq \mathbb{R}^d$ be a subspace with $\dim(V) = n$. The following is a basis set for V :

$$\{v_1, v_2, \dots, v_n\}$$

Let $b \in \mathbb{R}^d$.

GOAL : Project b onto V .

Let us define a matrix A as follows

$$A = \begin{bmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & \cdots & | \end{bmatrix}$$

The columns of A are linearly independent.

Let p be the projection of b onto V . Since $p \in V$, we can write $p = Ax$, where $x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$.

Let $e = b - p$ be the error vector. Since error vector is orthogonal to V , we can say that $A^T e = 0$.

$$\begin{bmatrix} - & v_1 & - \\ - & v_2 & - \\ & \vdots & \\ - & v_n & - \end{bmatrix} \times \begin{bmatrix} | \\ e \\ | \end{bmatrix} = \begin{bmatrix} v_1 \cdot e \\ v_2 \cdot e \\ \vdots \\ v_n \cdot e \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Let us expand $A^T e = 0$

$$\begin{aligned} A^T e &= 0 \\ A^T (b - p) &= 0 \\ A^T (b - Ax) &= 0 \\ A^T b - A^T Ax &= 0 \\ A^T Ax &= A^T b \\ x &= (A^T A)^{-1} A^T b \end{aligned}$$

Since the columns of A are linearly independent, $N(A) = \{0\}$. This means, $N(A^T A) = \{0\}$. Therefore, $A^T A$ is invertible.

The projection p is given by:

$$\begin{aligned} p &= Ax \\ &= A(A^T A)^{-1} A^T b \\ &= Pb \end{aligned}$$

Here, $P = A(A^T A)^{-1} A^T$ is referred to as the **projection matrix**.

Properties of Projection Matrix

Property-1 : P is a symmetric matrix. That is, $P^T = P$.

Proof:

$$\begin{aligned}
P^T &= \left(A(A^T A)^{-1} A^T \right)^T \\
&= (A^T)^T \left((A^T A)^{-1} \right)^T A^T \\
&= A(A^T A)^{-1} A^T
\end{aligned}$$

For the above proof, we have made use of the following:

- $(ABC)^T = C^T B^T A^T$
- $A^T A$ is symmetric, therefore, $(A^T A)^{-1}$ is also symmetric. That is, $\left((A^T A)^{-1} \right)^T = (A^T A)^{-1}$.

Property-2 : $P^2 = P$

Proof:

$$\begin{aligned}
P^2 &= P \times P \\
&= \left(A(A^T A)^{-1} A^T \right) \left(A(A^T A)^{-1} A^T \right) \\
&= A(A^T A)^{-1} (A^T A) (A^T A)^{-1} A^T \\
&= A(A^T A)^{-1} A^T
\end{aligned}$$

For the above proof, we have made use of the following:

- $(AB)C = A(BC)$
- $(A^T A)^{-1} (A^T A) = I$, where I is the identity matrix.

What does it mean to say that $P^2 = P$?

Suppose we are projecting a vector b onto V . The projection is Pb . What happens if we project Pb onto V ? Notice that $Pb \in V$. Projecting a vector onto a subspace in which it is already a part, results in the same vector. The projection of Pb onto V is given by $P \times Pb = P^2 b = Pb$.
